Run report: N83 precomputed-table rho control, run 1 (38855595549)

Workflow `.github/workflows/n83-precomputed-rho-control.yml`, commit `2f7d8c02`, GitHub Actions run 38855595549, started 2026-10-16T05:24 UTC. Two jobs (k3300, k600), each on one hosted `ubuntu-24.04` runner (4 vCPU, shared, L0). Arms per job: IC (frozen producer, one process, 16 targets), empty-table fold rho (16 processes), precomputed rho (a) 650 s and (b) = IC max RSS; IC summaries re-analysed in run 2 with the fixed analyzer. Targets: the 16 public points of `gha-38597929540-1`. Requested 16 targets per arm per job; completed 16 per arm per job, no failures or caps. Raw records: `records/run-38855595549/`. Protocol: `PROTOCOL.md`. Observations only; the verdict is in `REPORT.pdf`.

run 1 (38855595549) — IC K600 job (AMD EPYC 7763 64-Core Processor)

All recovered scalars replayed with the generic library curve and agreeing across arms and with the hosted answers: **`true`** (16 targets).

arm	median ms	95% CI	mean ms	95% CI
IC (frozen producer)	580.276	[168.726, 1340.436]	2024.844	[607.602, 3675.26]
rho, empty table (fold, Tier 1 worker)	1969.815	[1324.54, 3572.084]	2375.182	[1738.386, 3056.715]
rho + table, (a) 650 s, seed …601	3.333	[2.025, 4.7]	4.17	[2.721, 6.003]
rho + table, (b) = IC max RSS	1.471	[1.111, 2.218]	2.148	[1.34, 3.159]

paired ratio	median of ratios	95% CI	ratio of means	95% CI	num. slower
ic / bl_a_time_matched	214.73	[47.94, 483.84]	485.57	[131.91, 1072.5]	16/16
ic / bl_b_memory_matched	324.88	[148.1, 936.24]	942.57	[258.37, 2226.11]	16/16
rho_plain_fold / bl_a_time_matched	806.03	[399, 1148.52]	569.58	[333.1, 928.35]	16/16
rho_plain_fold / ic	5.25	[1.26, 6.32]	1.17	[0.58, 3.97]	12/16

precomputation	wall s	CPU s	memory	notes
IC	784.7	816.40	5.75 GB max RSS	rank threads 1
rho + table, (a) 650 s, seed …601	650.6	651	0.304 GB table	18987498 entries, 3.72 M steps/s, stop: wall_seconds
rho + table, (b) = IC max RSS	2818.3	11203	5.67 GB table	354371105 entries, 8.07 M steps/s, stop: table_bytes

run 1 (38855595549) — IC K3300 job (AMD EPYC 9V45 96-Core Processor)

All recovered scalars replayed with the generic library curve and agreeing across arms and with the hosted answers: **`true`** (16 targets).

arm	median ms	95% CI	mean ms	95% CI
IC (frozen producer)	24.774	[18.425, 50.21]	37.278	[23.802, 53.249]
rho, empty table (fold, Tier 1 worker)	1361.144	[914.186, 2456.497]	1636.179	[1200.606, 2108.434]
rho + table, (a) 650 s, seed …601	2.321	[1.4, 2.797]	2.33	[1.76, 2.968]
rho + table, (b) = IC max RSS	0.841	[0.391, 1.155]	0.86	[0.609, 1.14]

paired ratio	median of ratios	95% CI	ratio of means	95% CI	num. slower
ic / bl_a_time_matched	16.26	[7.23, 22.6]	16	[9.23, 26.77]	15/16
ic / bl_b_memory_matched	42.72	[25.49, 74.91]	43.34	[25.79, 75.05]	15/16
rho_plain_fold / bl_a_time_matched	805.56	[399.17, 1147.27]	702.26	[484.03, 963.21]	16/16
rho_plain_fold / ic	63.45	[17.66, 85.23]	43.89	[24.22, 80.21]	16/16

precomputation	wall s	CPU s	memory	notes
IC	497.6	603.71	10.79 GB max RSS	rank threads 4
rho + table, (a) 650 s, seed …601	650.6	651	0.455 GB table	28434341 entries, 5.58 M steps/s, stop: wall_seconds
rho + table, (b) = IC max RSS	3126.3	12429	10.519 GB table	657460555 entries, 13.67 M steps/s, stop: table_bytes

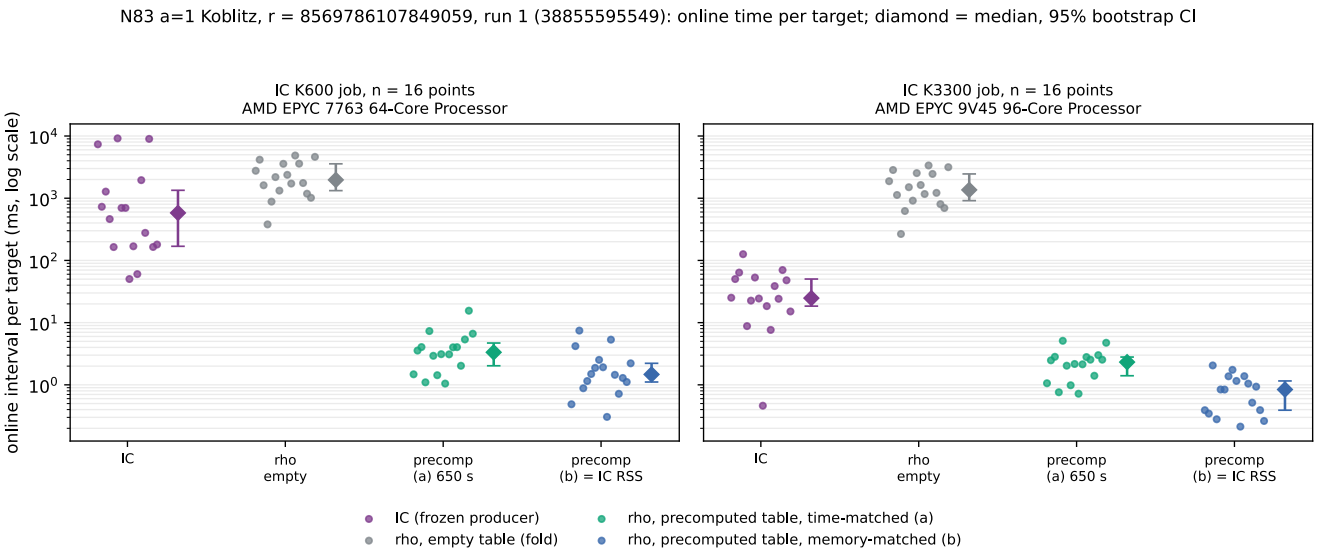


Figure 1: run 1 (38855595549): online interval per target; diamond = median with 95% bootstrap interval.